

Product Initial Verification Test Plan

Project: ESP32 + DRV8825 Development Kit

Document ID: P1012202501-Test_Plan.md **Owner:** Project Design Lead

Version: 1.0

Date: 2025-12-15

Purpose and Scope

Purpose: Define the testing requirements designed to validate the functional, performance, safety, and regulatory requirements for a modular development kit integrating an ESP32 micro-controller and a DRV8825 stepper motor driver.

Scope: Applies to the ESP32 based DRV8825 stepper motor driver with RS-485 interface.

Required Materials

- **P1012202501-01 Test Plan Log.xlsx** file located in **/ProjectDocuments/test-logs/** to store the results of the testing process. Use 1 test plan log file per PCB serial number tested.
 - Schematic and BOM for the device.
 - Fully assembled device(s) for testing.
 - Bench top power supply capable of current limiting with the ability to deliver up to 30V @ 5A.
 - Multi-Meter capable of resistance, voltage, current and temperature reading.
 - Basic hand tools such as screwdrivers.
 - Computer system with USB connectors and software to program the ESP32.
 - Appropriate USB-C cable with power and data to connect to the PCB assembly.
 - Terminal emulator software on PC capable of communication over serial UART interfaces. An example would be PuTTY.
 - USB to UART interface cable such as an RS-232 FTDI assembly.
 - USB to RS-485 adapter and two small lengths of wire for interfacing to the RS-485.
 - Assembled and tested momentary pushbutton for limit and control input testing.
 - Click PLC **pn here** with testing program **P1012202501InputTest** installed.
 - The Click PLC should have the **pn here** discreet input/output module installed and working.
 - Appropriate 4 wire, bipolar stepper motor.
 - Desktop thermometer capable of reading °C ambient temperature values.
 - Hot air gun with temperature control.
-

Prior to Testing

- Open a fresh copy of **P1012202501-01 Test Plan Log.xlsx** and then save it as **P1012202501-01 Test Plan Log xxxxxxxx.xlsx** where xxxxxxxx is the serial number of the board that the test log is associated with in the **\ProjectDocuments\test-logs** folder. Ensure that the measurement equipment is recorded in the test plan log.
-

For any test, note the reason for a fail condition and stop testing to investigate the issue and note any resolutions

Initial Testing

Visual Inspection

- Test 1: Visually inspect the assembled board for any workmanship defects such as poor soldering or off-pad components following IPC Class II criteria. Store the results in the log file.

Functional Requirements

These tests are verification only and are performed during the visual inspection. Refer the design schematics for additional verifications.

- FR-001: Does the design include an ESP32 with Wi-Fi and BLE capabilities?
- FR-002: Does the design include the Texas Instruments DRV8825 stepper motor driver IC with step, direction, and enable inputs?
- FR-003: Does the design support 12-24VDC input with integrated 3.3V and 5.0V regulation for logic circuits?
- FR-004: Does the design incorporate pluggable headers for the RS-485, control inputs, limit inputs, and motor outputs?
- FR-005: Does the design incorporate a USB capable interface for programming the ESP32?
- FR-006: Does the design incorporate test points for the power rails?
- FR-007: Does the design incorporate LEDs for visual indication of function or status?

Initial PCB Assembly Testing

Initial resistance measurements

- Test 2: Using the multi-meter in resistance mode, check between TP1 (+3.3V) and TP 3 (GND) and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 3: Using the multi-meter in resistance mode, check between TP2 (+5.0V) and TP 3 (GND) and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 4: Using the multi-meter in resistance mode, check between TP4 (RS485 RX) and TP 7 (GND) and verify that the resistance is greater than 1K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 5: Using the multi-meter in resistance mode, check between TP5 (RS485 DIR) and TP 7 (GND) and verify that the resistance is greater than 1K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 6: Using the multi-meter in resistance mode, check between TP6 (RS485 TX) and TP 7 (GND) and verify that the resistance is greater than 1K Ω . If the measured value is below, stop and inspect the

board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.

- Test 7: Using the multi-meter in resistance mode, check between TP4 (RS485 RX) and TP 5 (RS485 DIR) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 8: Using the multi-meter in resistance mode, check between TP4 (RS485 RX) and TP 6 (RS485 TX) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 9: Using the multi-meter in resistance mode, check between TP5 (RS485 DIR) and TP 6 (RS485 TX) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 10: Using the multi-meter in resistance mode, ensure that J1 does not have a jumper installed then check between CN2 Terminal 1 and Terminal 2 and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 11: Using the multi-meter in resistance mode, install a jumper on J1 and then check between CN2 Terminal 1 and Terminal 2 and verify that the resistance is around 120Ω. If the measured value is greater or less than 50Ω, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 12: Using the multi-meter in resistance mode, check between TP10 (Limit 1) and TP13 (GND) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 13: Using the multi-meter in resistance mode, check between TP11 (Limit 2) and TP13 (GND) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 14: Using the multi-meter in resistance mode, check between TP12 (Limit 3) and TP13 (GND) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 15: Using the multi-meter in resistance mode, check between TP10 (Limit 1) and TP11 (Limit 2) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 16: Using the multi-meter in resistance mode, check between TP10 (Limit 1) and TP13 (Limit 3) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 17: Using the multi-meter in resistance mode, check between TP12 (Limit 2) and TP13 (Limit 3) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the

board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.

- Test 18: Using the multi-meter in resistance mode, ensure that J2 does not have a jumper installed then check between CN3 Terminal 1 and Terminal 4 and verify that the resistance is greater than 1K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 19: Using the multi-meter in resistance mode, ensure that J2 does not have a jumper installed then check between CN3 Terminal 2 and Terminal 4 and verify that the resistance is greater than 1K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 20: Using the multi-meter in resistance mode, ensure that J2 does not have a jumper installed then check between CN3 Terminal 3 and Terminal 4 and verify that the resistance is greater than 1K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 21: Using the multi-meter in resistance mode, check between CN3 Terminal 1 and Terminal 2 and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 22: Using the multi-meter in resistance mode, check between CN3 Terminal 1 and Terminal 3 and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 23: Using the multi-meter in resistance mode, check between CN3 Terminal 2 and Terminal 3 and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 24: Using the multi-meter in resistance mode, check between CN4 Terminal 1 and Terminal 4 and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 25: Using the multi-meter in resistance mode, check between CN4 Terminal 2 and Terminal 4 and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 26: Using the multi-meter in resistance mode, check between CN4 Terminal 3 and Terminal 4 and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 27: Using the multi-meter in resistance mode, check between CN4 Terminal 1 and Terminal 2 and verify that the resistance is greater than 5K Ω . If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.

- Test 28: Using the multi-meter in resistance mode, check between CN4 Terminal 1 and Terminal 3 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 29: Using the multi-meter in resistance mode, check between CN4 Terminal 2 and Terminal 3 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 30: Using the multi-meter in resistance mode, check between TP24 (VBUS) and TP25 (GND) and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 31: Using the multi-meter in resistance mode, check between TP22 (USB DP) and TP23 (USB DN) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 32: Using the multi-meter in resistance mode, check between TP18 (RTS) and TP19 (DTR) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 33: Using the multi-meter in resistance mode, check between TP20 (BOOT) and TP21 (ENABLE) and verify that the resistance is greater than 1KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 34: Using the multi-meter in resistance mode, check between CN7 Terminal 1 and TP7 (GND) and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 35: Using the multi-meter in resistance mode, check between CN7 Terminal 2 and TP7 (GND) and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 36: Using the multi-meter in resistance mode, check between CN7 Terminal 3 and TP7 (GND) and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 37: Using the multi-meter in resistance mode, check between CN7 Terminal 4 and TP7 (GND) and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 38: Using the multi-meter in resistance mode, check between CN7 Terminal 1 and Terminal 4 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 39: Using the multi-meter in resistance mode, check between CN7 Terminal 2 and Terminal 4 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the

board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.

- Test 40: Using the multi-meter in resistance mode, check between CN7 Terminal 3 and Terminal 4 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 41: Using the multi-meter in resistance mode, check between CN7 Terminal 1 and Terminal 3 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 42: Using the multi-meter in resistance mode, check between CN7 Terminal 2 and Terminal 3 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.
- Test 43: Using the multi-meter in resistance mode, check between CN7 Terminal 1 and Terminal 2 and verify that the resistance is greater than 5KΩ. If the measured value is below, stop and inspect the board for assembly issues that could cause a short or low resistance reading such as a solder bridge. If problem is not identifiable, elevate to direct supervisor for next steps.

Initial board power up testing

- Test 44: Set the bench DC power supply to 5V with a current limit of 100mA. Ensure that the power output is disabled. Connect the red positive output to CN1 Terminal 1 and the black negative output to CN1 Terminal 2. Enable the power supply output and check if the current limit trips. Also observe the assembled PCB for any electrical issues such as arcs or smoke. If any of the failure conditions arise, disable the power supply output and inspect the assembly for issues that could cause these issues. If the problem is not identifiable, elevate to direct supervisor for next steps.
- Test 45: Disable the power supply output. Set the output voltage to 24VDC and the current limit to 500mA. Enable the power supply output and check if the current limit trips. Also observe the assembled PCB for any electrical issues such as arcs or smoke. If any of the failure conditions arise, disable the power supply output and inspect the assembly for issues that could cause these issues. If the problem is not identifiable, elevate to direct supervisor for next steps.
- Test 46: With the power supply output enabled, configure a multimeter to DC voltage measurement and connect the negative lead to TP3 (GND) and the positive lead to TP1 (3.3V). Check and verify that the voltage output is positive 3.3V +/- 125mV. If any failure conditions arise, disable the power supply output and inspect the assembly for issues that could cause these issues. If the problem is not identifiable, elevate to direct supervisor for next steps.
- Test 47: With the power supply output enabled, configure a multimeter to DC voltage measurement and connect the negative lead to TP3 (GND) and the positive lead to TP2 (5.0V). Check and verify that the voltage output is positive 5.0V +/- 125mV. If any failure conditions arise, disable the power supply output and inspect the assembly for issues that could cause these issues. If the problem is not identifiable, elevate to direct supervisor for next steps.

- SST 1: With the PCB assembly connected to the power supply set at 24VDC and the output enabled, connect the USB-C cable to the computer and the PCB assembly. Does the system recognize a newly connected serial port? This validates that the USB to UART bridge chip is operating properly and that the computer does detect an operational COM port.
 - SST 2: Using the computer and ESP32 programming software, flash the **HelloWorld** firmware and validate that the system can program the ESP32 and that basic communication does happen over the UART interface. This test validates that the ESP32 can be programmed, boots and operates normally. May need to use the BOOT and ENABLE buttons on the PCB for the first ESP32 flashing.
 - SST 3: Using the computer and ESP32 programming software, flash the **UARTFTDI** firmware. Connect the USB to FTDI adapter to the PC and to the PCB at J4 observing proper connector orientation. Load up PuTTY with these settings for the appropriate COM port, baud rate of 115200, 8 data bits, 1 stop bit and no parity. Upon reset or after pressing the EN button, this message should appear on the terminal: P1012202501 UART FTDI Interface Test Firmware.
-

In the terminal window, type in the following: echo test A response should be sent back that looks like the following: Echo: test If the response is successful, this is a pass condition and verifies that the UART interface is working properly.

- SST 4: Using the computer and ESP32 programming software, flash the **STATUSLD9** firmware to the ESP32. Observe the red LED D9 and verify that it is flashing at a rate of about 5 second intervals.
- SST 5: Using the computer and ESP32 programming software, flash the **RS485TEST** firmware. Connect the USB to RS-485 adapter to the PC and to the PCB at CN2 observing proper connections to A and B. Load up PuTTY and validate that basic bi-directional communication does happen over the UART FTDI (J4) interface.
- SST 6: Using the computer and ESP32 programming software, flash the **INPUTTEST** firmware. Install 1 jumper onto J2 and 1 jumper onto J3. This will connect the common circuits to the internal ground plane. Connect the test button unit wire #1 to the 24VDC power supply positive connection. Connect the #2 wire to CN3 terminal #1. Check the status LED D9 to see if it is flashing. If it is not, press and hold the test button and observe if D9 is indicating a single flash with a delay of 2 seconds. If this test is successful, disconnect the #2 wire from CN3.
- SST 7: Connect the #2 wire to CN3 terminal #2. Check the status LED D9 to see if it is flashing. If it is not, press and hold the test button and observe if D9 is indicating a double flash with an interval of 1 second and a delay of 2 seconds. If this test is successful, disconnect the #2 wire from CN3.
- SST 8: Connect the #2 wire to CN3 terminal #3. Check the status LED D9 to see if it is flashing. If it is not, press and hold the test button and observe if D9 is indicating a triple flash with an interval of 1 second and a delay of 2 seconds. If this test is successful, disconnect the #2 wire from CN3.
- SST 9: Connect the #2 wire to CN4 terminal #1. Check the status LED D9 to see if it is flashing. If it is not, press and hold the test button and observe if D9 is indicating a quadruple flash with an interval of 1 second and a delay of 2 seconds. If this test is successful, disconnect the #2 wire from CN4.
- SST 10: Connect the #2 wire to CN4 terminal #2. Check the status LED D9 to see if it is flashing. If it is not, press and hold the test button and observe if D9 is indicating a series of five flashes with an interval of 1 second and a delay of 2 seconds. If this test is successful, disconnect the #2 wire from CN4.
- SST 11: Connect the #2 wire to CN4 terminal #3. Check the status LED D9 to see if it is flashing. If it is not, press and hold the test button and observe if D9 is indicating a series of six flashes with an interval of 1 second and a delay of 2 seconds. If this test is successful, disconnect the #2 wire from CN4.

- SST 12: With the **INPUTTEST** firmware flashed to the PCB assembly, remove the jumpers on J2 and J3. Disconnect any wires presently tied to CN3 or CN4. Connect the test button #1 wire to the PLC power supply positive terminal block. Connect the #2 wire to the PLC X1 terminal. Connect the PCB assembly CN3 terminal 1 to the PLC Y1 output, the CN3 terminal 2 to Y2 output, the CN3 terminal to Y3 output and the CN3 terminal 4 to the PLC positive power supply. Connect the PCB assembly CN4 terminal 1 to the PLC Y4 output, the CN4 terminal 2 to the PLC Y5 output, the CN4 terminal 3 to the PLC Y6 output, and the CN4 terminal 4 to the PLC positive power supply. Verify that 24VDC is applied to the PCB assembly and that the PLC is also powered up properly. Place the PLC into **RUN** mode and wait a few seconds for it to settle. Press the test button and observe the PCB assembly status LED D9. The LED should follow a similar blink pattern as in the prior test but in sequence as each input is tested. Once testing is completed, set the PLC to **STOP** mode and power down. Power down the PCB assembly and disconnect from the PLC for next test phase.
- SST 13: Ensure that the bench top power supply is set to 24VDC and the current limit is set to at least 2A. Flash the PCB assembly with the **STEPPERTEST** firmware. Disable the power supply output then connect the motor A+ input to CN7 terminal 1, connect the A- input to CN7 terminal 2, connect the B+ input to CN7 terminal 4, and the B- input to CN7 terminal 3. Refer to the stepper motor documentation for proper input identification. Enable the power supply output and observe the stepper motor. It should rotate clockwise for about 30 seconds then stop and start rotating counter-clockwise for another 30 seconds then stop and repeat the cycle. If this is observed, the test is a success.
- SST 14: With the bench top power supply enabled, flash the **FANTEST** firmware to the ESP32 and then disable the bench top power supply output and connect a **FANDC5V40x40 Assembly** to CN6 and then re-enable the power output of the supply. The fan should not be operating. Connect the USB to FTDI converter to J4 observing proper polarity. Load up PuTTY or a similar terminal emulator application and connect to the ESP32 serial console with the following settings: Baud rate 115200, 8 data bits, 1 stop, no parity. Reset the PCB assembly by pressing and releasing the **ENABLE** button. A message should appear on the terminal screen that reads **P1012202501 Assembly Fan Test**. After this message, type in the command **getspeed** (all one word). The system should return 0 indicating that the fan is not rotating as it should not be. Type in the command and argument **setspeed 30**. The fan should start rotating at this point at a slow speed. Type in the command **getspeed** and the system should return a value in RPM. This value will fluctuate depending on the tolerance of the detection circuitry in the fan and other variables that are outside of our control. Observe the fan for 1 minute to verify that it will continually operate properly. Type in the command **setspeed 100** and observe if the fan speeds up. Type in the command **getspeed** and the system should return an RPM value higher than the prior speed level. Observe the fan for 1 minute to verify proper operation. Type in the command **setspeed 0** and observe if the fan stops rotating. Type in the command **getspeed** and the returned value should be 0. Observe the fan for 1 minute to verify that it does not start rotating without any input. Record the results of this test into the log file. Disable the bench top power supply and disconnect the fan assembly from CN6.
- SST 15: Connect the USB to FTDI cable to J4 observing proper polarity and then enable the power supply output. Flash the **TEMPTEST** firmware to the ESP32. Load up PuTTY or similar terminal emulator and load the following settings on the proper COM port, 115200 Baud Rate, 8 data bits, 1 stop bit and no parity. Reset the PCB assembly by pressing and releasing the **ENABLE** button and observe the terminal software for the following message: **P1012202501 Temperature Sensor Test**. At this point, type the command **gettemp** and press enter, the system should return the current measured temperature in °C that is closely aligned with a desktop thermometer within a few degrees. Set the temperature of the hot air gun to a low setting **that does not burn you** and apply the hot air to the

PCB temperature sensor IC7 and hold for 30 seconds. Turn off the hot air gun and then type the command `gettemp` and the ESP32 should return with a temperature value higher than the prior test. Allow the PCB assembly to cool for 5 minutes and then type the command `gettemp` and the ESP32 should respond with a value closer to desktop thermometer.